

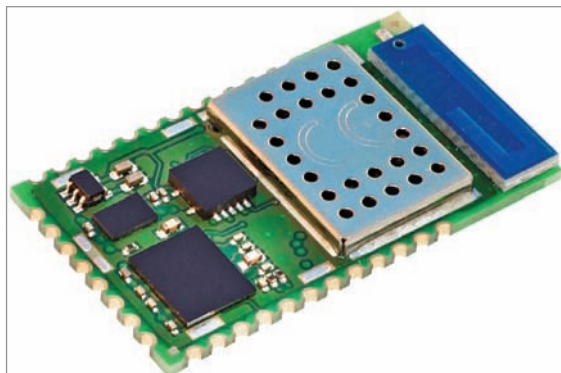


Fig. 2: SPWF04 Wi-Fi module (Credit: <https://blog.st.com>)



Fig. 3: ESP pre-certified modules with antenna and without antenna

follow all the regulatory guidance from its datasheet, then you can avoid certification in the final design because the company already has taken care of this.

These modules are created by manufacturers that have completed all necessary steps for obtaining FCC certification. By incorporating these modules into your design, you gain the advantage of complying with FCC regulations. However, this choice comes with limitations on antenna selection. But there is a way for that as well.

The original equipment manufacturer (OEM) responsible for these modules collaborates with an EMC testing laboratory to certify a set of antennas compatible with the transmitter. As a result, you, as the end user, must select an identical antenna to those used for FCC compliance. Most pre-certified modules are accompanied by an integration manual that outlines how to seamlessly incorporate the module into

your existing design project.

The design trap

While selecting the components there can be an issue as most companies often launch these in series with slight differences. One slight mistake in design and selection can void your FCC and CE certification and cost a large penalty

for your company.

Here is an example of one mistake most budding design engineers might unknowingly make.

As you can see in Fig. 3, there are two components of ESP, and both are FCC and CE certified, both have the same specs, but there is a difference. If I use the first with an antenna in my design, then there's a higher chance that my product design doesn't need FCC certification. If I select the second chip for the design, then it has no onboard antenna; instead, it has the connector pin for that. If I add an onboard antenna to the design, the FCC certification of the overall design gets void, and I need to recertify the design for the whole board. But if I ship and sell the final design without the antenna and leave the user to get it separately and connect it as per their need, this can give relief to my design product from certification. However, the end customer who is going to use it needs to take care of certification for the design after adding the antenna in their deployed design. In such a case, if your customer is a company (B2B), then the company using your design product needs to recertify the product if they add the antenna to it. **EFY**

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-Jeff Bezos

